User Flow: Core Happy-Path Journey

The exact 6-step journey used in the live demo (primary actor: Agriculture Extension Officer; end recipient: farmer)

> **Demo scenario:** It is a Monday in July. IMD's block forecast for **Gajwel block, Siddipet district** shows a uniform "light rain, 8 mm". PanchayatCast reveals that some panchayats will get heavy rain while others stay dry - and that changes what cotton farmers should do this week. (In Telangana, blocks correspond to mandals. Village/GP names here are illustrative; replace them with real GPs from your pilot dataset.)

A. Numbered Flow

1. **Select region** [Screen: Region Selection / Home]

   AEO Lakshmi opens PanchayatCast and selects **Telangana > Siddipet > Gajwel**. The app loads today's IMD block forecast and the downscaled panchayat layer (already computed by the daily pipeline).

2. **Compare block vs panchayat** [Screen: Block vs Panchayat Map]

   A split map shows the flat block forecast (left) next to the panchayat-level map (right). She switches variable (Rain / Tmax / Tmin / RH / Wind) and day (Day 1-5) to see local variation.

3. **Drill into a panchayat** [Screen: Panchayat Forecast Detail]

   She taps **Ahmadipur GP**. A detail panel shows the 5-day forecast cards, rain probability, the block value for comparison, and a **High / Medium / Low confidence** badge.

4. **Generate crop advisory** [Screen: Crop Advisory Generator]

   She chooses crop **Cotton** and stage **Square formation**. The rule engine returns prioritised advisories, e.g. "Postpone pesticide spray: 72% chance of >10 mm rain in 48 h".

5. **Translate & send** [Screen: Share & Farmer Preview]

   She previews the advisory in **Telugu** (also Hindi / English) and taps **Send to Ahmadipur farmers**. A phone mockup shows the SMS / WhatsApp message arriving for farmer Ramesh.

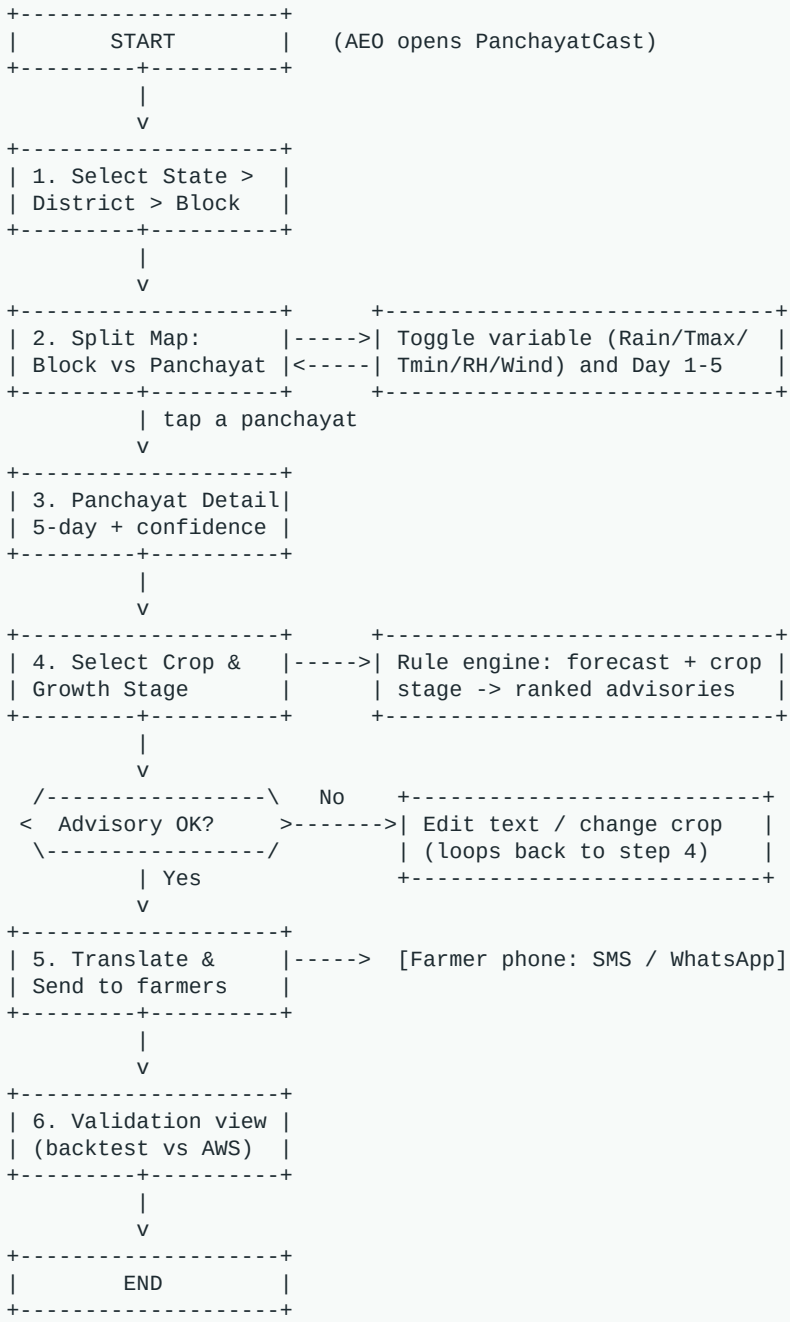
6. **Prove accuracy** [Screen: Model Validation & Accuracy]

   She opens the validation view: on a past event day, the panchayat forecast matched the nearby AWS reading while the block value missed it; overall error vs the block baseline is displayed.

Step Summary

#	User action	System response	Must feature shown	Time
1	Selects district & block	Loads block forecast + downscaled layers	M1 Downscaling engine	~10 s
2	Toggles variable / day	Split map re-renders instantly	M2 Comparison map	~15 s
3	Taps a panchayat	5-day cards + confidence badge	M1, M4 Confidence	~15 s
4	Picks crop & stage	Actionable advisories generated	M3 Advisory engine	~15 s
5	Previews Telugu, taps Send	Message appears on phone mockup	M5 Local-language delivery	~15 s
6	Opens validation	Backtest vs AWS + error metrics	M4 Validation	~15 s

B. Visual Flowchart



C. Drawing Guide (for Figma / draw.io / Excalidraw)

Node	Shape	Label	Colour	Connects to
N0	Rounded pill	Start: AEO opens app	Dark green #0B5D4B	N1
N1	Rectangle	Select State > District > Block	Light green #EAF4F1	N2
N2	Rectangle	Split map: Block vs Panchayat	Light green	N2a (loop), N3
N2a	Side note box	Toggle variable & day	White, dashed border	back to N2
N3	Rectangle	Panchayat detail: 5-day + confidence	Light green	N4
N4	Rectangle	Select crop & growth stage	Light green	N4a, D1
N4a	Side note box	Rule engine generates advisories	Amber #E8A317 border	-
D1	Diamond	Advisory OK?	Amber	Yes -> N5; No -> N4b
N4b	Rectangle	Edit text / change crop	White	back to N4
N5	Rectangle	Translate & send to farmers	Light green	N5a, N6
N5a	Phone icon	Farmer receives SMS / WhatsApp	Amber	-
N6	Rectangle	Validation view (backtest vs AWS)	Light green	N7
N7	Rounded pill	End	Dark green	-

Layout tips

- Draw the main path top-to-bottom in one column; place side notes (N2a, N4a, N5a) to the right.
- Put the step number inside every main box so it matches the demo script and screen list.
- Keep the decision loop (D1 -> N4b) small; in the live demo always take the "Yes" path.
- Use the same colours as the product UI so the flowchart slide feels consistent with the demo.

Fallback path: if the network fails during judging, the app runs in **Demo Mode** with a cached forecast date, so every step above still works offline.